

Design and Development of an Automated High-Magnification Alt-Azimuth Telescope System

ME-366 Final Project Report

Group B-3

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Abstract

This report details the design, mechanical fabrication, and control system integration of an automated, high-magnification Alt-Azimuth (Alt-Az) astronomical refractor telescope. The system implements a hierarchical multi-tier control architecture comprising a high-level Python backend control hub integrated with Stellarium planetarium software for celestial targeting and coordinate transformations, an intermediate ESP32 microcontroller for real-time stepping pulse synthesis, and dedicated motor drivers supplying 1:16 microstepping to NEMA 17 stepper motors. The mechanical drive employs a custom 120:1 worm gear reduction mechanism on both axes, yielding an ultra-fine angular tracking resolution of approximately 3.375 arcseconds per step while preventing gravitational back-driving. In addition, an integrated camera control module with exposure control and intervalometer functions enables automated celestial capture. The project is evaluated against Complex Engineering Problem (CEP) criteria, followed by a discussion of fabrication challenges, worm drive redesign iterations, and future enhancements.

1. Introduction

Astronomical instrumentation demands high angular precision, rigid structural support, and synchronized optical-mechanical control. For the ME-366 course, we designed and fabricated a motorized Alt-Azimuth refractor telescope system tailored for visual observation and digital photography.

The project documentation, hardware design files, and source code are maintained in an open repository: `M000nwalker/Beyond_Horizon_1`.

2. System Architecture & Key Features

2.1 Hierarchical System Architecture

The automated telescope relies on a three-tier distributed control and actuation architecture designed to decouple high-level astronomical computation from low-level, time-critical motor pulse generation (Figure 1):

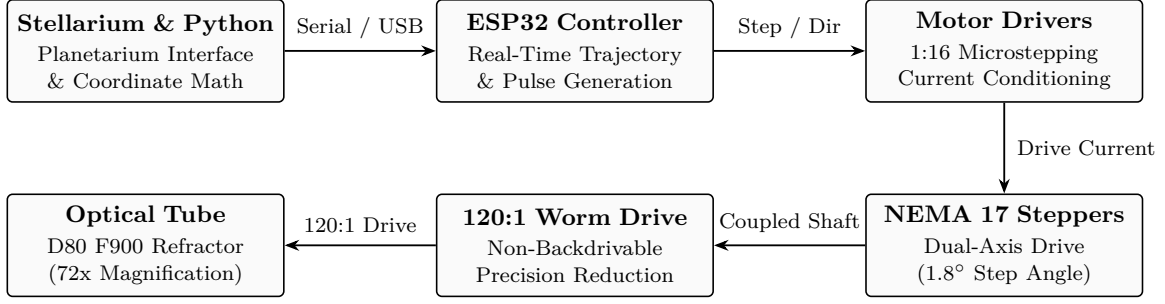


Figure 1: Three-tier system architecture from Stellarium and Python control hub to physical actuation.

1. **Tier 1 — Stellarium & Python Backend Hub:** Operates on a host PC/laptop. Observers select celestial targets visually within the built interface. A Python backend script interfaces with Stellarium (via the Stellarium Telescope Server Protocol), retrieves celestial coordinates (Right Ascension α , Declination δ), transforms them in real time into local horizontal coordinates (Altitude h , Azimuth A) using geographic location and system time, and streams directional step vectors to the microcontroller.
2. **Tier 2 — ESP32 Microcontroller Layer:** An ESP32 dual-core SoC handles real-time execution. It receives target coordinate packets over a dedicated communication link, executes acceleration/deceleration trapezoidal profiling, and generates deterministic, jitter-free step and direction pulses.
3. **Tier 3 — Motor Driver & Actuation Stage:** Dedicated bipolar stepper drivers receive pulse trains and condition phase currents at 1:16 microstepping, driving dual NEMA 17 stepper motors coupled to 120:1 worm gear reducers.

2.2 120:1 Worm Drive & 1:16 Microstepping Actuation

Astronomical tracking under high magnification requires smooth, low-velocity motion and high mechanical holding torque:

- **NEMA 17 Stepper Motors:** Bipolar NEMA 17 stepper motors (1.8° step angle, 200 full steps per revolution) provide ample holding torque to stabilize the cantilevered moment of the long optical tube.
- **1:16 Microstepping:** Motor driver channels are configured for 1:16 microstepping, yielding $200 \times 16 = 3,200$ microsteps per motor revolution. This eliminates low-speed stepping resonance, prevents optical vibration, and ensures smooth continuous motion during observation.
- **120:1 Worm Gear Reduction:** A custom 120:1 worm-and-wheel reduction drive is deployed on both the Altitude and Azimuth axes. In addition to delivering a 120-fold mechanical torque multiplication, the worm drive is naturally non-backdrivable, preventing the optical tube from tilting or slipping under gravitational imbalance when unpowered.
- **High-Resolution Kinematics:** The combined drivetrain produces an exceptional angular resolution:

$$\text{Microsteps per Axis Revolution} = 200 \times 16 \times 120 = 384,000 \text{ steps/rev} \quad (1)$$

$$\text{Angular Resolution} = \frac{360^\circ}{384,000} \approx 0.0009375^\circ \approx 3.375 \text{ arcseconds/step} \quad (2)$$

This fine resolution ensures smooth celestial tracking without observable stepped jumps in the field of view at 72x magnification.

2.3 D80 F900 Optical Refractor Assembly

The optical tube assembly utilizes a **D80 F900** achromatic objective lens (80 mm clear aperture, 900 mm focal length). This optical configuration provides significant performance advantages:

- **High Focal Ratio ($f/11.25$):** The long 900 mm focal length relative to the 80 mm aperture severely reduces chromatic aberration and spherical distortion without requiring complex corrective glass groups.
- **Superior Visual Contrast:** The clean optical path produces sharp, high-contrast images, ideal for observing planetary surface details and lunar topography.
- **72x Maximum Zoom:** Configured to achieve a maximum usable magnification of **72x**, allowing the system to resolve fine celestial targets while retaining optical sharpness.

2.4 Stellarium Planetarium Integration & Python Control Hub

Rather than requiring manual coordinate lookups or complex numeric command entries, the telescope integrates with **Stellarium**, a widely used open-source planetarium platform:

- **Stellarium and Control Hub Communication:** A Python background script acts as a bridge, communicating with Stellarium's database upon user request for celestial co-ordinates.
- **Coordinate Transformation:** The Python hub automatically converts Stellarium's celestial RA and Declination coordinates into local horizontal Altitude (Alt) and Azimuth (Az) angles based on time and geographic coordinates.
- **Automated Alignment & Slew:** Step commands are dispatched through the ESP32 to align the optical axis with the target and maintain tracking compensation for Earth's rotation.

2.5 Camera Control Module with Intervalometer

To support astrophotography, the system incorporates a camera control module that provides:

- **Exposure Control:** Handles precise shutter timing for capturing faint objects.
- **Integrated Intervalometer:** Enables automated sequential frame capture at set intervals for time-lapse observations and frame stacking.

3. Justification of Complex Engineering Problem (CEP) Criteria

The project satisfies three core Complex Engineering Problem (CEP) criteria:

3.1 P-1: Depth of Knowledge

Execution required combining principles across multiple engineering domains:

- **Geometric Optics:** Analyzing focal ratios ($f/11.25$), diffraction limits, and chromatic performance for the D80 F900 refractor assembly.
- **Precision Mechanical Design:** Designing an Alt-Az mount capable of smooth motion while resisting deflection under cantilevered tube loads, integrating custom 120:1 worm gear sets for self-locking anti-backdrive performance.
- **Mechatronics & Embedded Systems:** Developing a multi-tier control hierarchy spanning Stellarium planetarium integration, Python ephemeris math, low-latency ESP32 pulse synthesis, and 1:16 microstepping current regulation for NEMA 17 stepper motors.

3.2 P-2: Conflicting Requirements

The engineering process required resolving several trade-offs:

- **Rigidity vs. Structural Weight:** High magnification (72x) amplifies micro-vibrations, demanding heavy structural support, whereas portable operation constrained total assembly mass.
- **Focal Length vs. Tracking Stability:** The long 900 mm tube creates a substantial moment arm, increasing wind resistance and requiring higher motor holding torque.

3.3 P-3: Depth of Analysis

Detailed analytical modeling was necessary to ensure system performance:

- **Drive Kinematics & Angular Resolution:** Analyzing gear reduction and microstepping to prove that a 120:1 worm drive coupled with 1:16 microstepping (384,000 microsteps/rev) delivers an ultra-fine resolution of $3.375''$ per step, well within the optical resolution threshold at 72x zoom.
- **Coordinate Transformation Math:** Formulating matrix transformations to convert celestial Right Ascension and Declination (α, δ) from Stellarium into instantaneous local horizontal coordinates (h, A) in real time.

4. Engineering Challenges & Failures

4.1 Worm Gear Manufacturing Limitations

The design specified custom 120:1 worm gear reduction drives for both axes to achieve fine resolution and intrinsic anti-backdrive functionality. Initial efforts to machine the high-ratio 120:1 worm gear assembly out of metal failed due to tooling constraints and tight thread pitch manufacturing tolerances, forcing a transition to additive manufacturing (3D printing).

4.2 Worm Screw Failure & Redesign Iterations

Transitioning to printed drive components introduced severe mechanical reliability challenges:

- **Worm Screw Shearing:** The primary 3D-printed **worm screw** experienced repeated mechanical shearing under driving torque and axial loads during high-speed slewing.
- **Material Deflection:** Polymer worm screws exhibited flexure, leading to tooth binding against the 120-tooth worm wheel and eventual fatigue fracture.
- **Iterative Redesign:** The team had to re-engineer the worm screw geometry over multiple design cycles, adjusting tooth pitch profiles, infill density, print orientation, and sleeve bearings before achieving structural stability.

4.3 Control Integration

Developing reliable software logic to synchronize celestial ephemeris updates on the Python host with real-time pulse streaming on the ESP32 without losing steps under load required extensive algorithm optimization and iterative testing.

5. Future Enhancements

To build upon the current working prototype, three main upgrades are planned:

1. **Dedicated Physical Camera Mount:** Manufacturing a rigid, vibration-damped camera mounting bracket to securely hold camera hardware at the focal plane for high-resolution lunar imaging.
2. **Ergonomic Vertical Focuser:** Designing a vertical right-angle focusing mechanism to enable comfortable viewing angles when the optical tube is pointed near the zenith.
3. **Cloud-Based Control Server Migration:** Offloading coordinate processing and database queries to a cloud server, allowing full control via mobile devices without needing a tethered laptop.

6. Conclusion

The ME-366 telescope project successfully demonstrated an automated Alt-Az refractor system combining a D80 F900 optical tube (72x magnification) with a multi-tier control architecture (Stellarium-integrated Python backend to ESP32 to NEMA 17 steppers). By implementing 1:16 microstepping and custom 120:1 worm reduction drives, the system achieved a sub-four-arcsecond ($3.375''$) tracking resolution with anti-backdrive stability. Despite early manufacturing setbacks in metal machining and repeated 3D-printed worm screw shearing, iterative engineering delivered a robust, functional automated astronomical telescope.